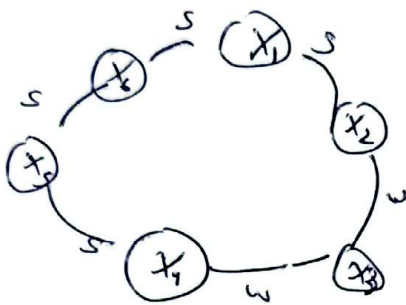


Question 1:

a) Variables = $x_1, x_2, x_3, x_4, x_5, x_6$.

Domain = (pit, ghost, exit).

Binary constraints as question says that exit are not adjacent
 $(x_i, x_j) \neq (Exit, Exit)$



s = means atleast one of the (x_i) or (x_j) is pit (Strong breeze).

w = both are not pit atleast one of them is an exit.

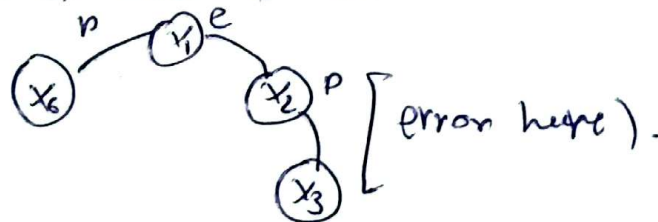
b) $x_1 = Exit \rightarrow$ Unary constraint

thus if we assign x_1 exit, the neighbors could not be exit as per constraint $x_2 \neq E$ and $x_6 \neq E$.

thus
 x_2 can be pit (Strong breeze)
 or x_6 can be a pit

Updating the domains of x_2 and x_6 .

$x_2 = \{s\}$ and $x_6 = \{s\}$.



In x_2 and x_3 pit Pacman stands in between he should experience weak breeze ~~with no~~ which means x_2 and x_3 should not be pit, in addition there is atleast one exit thus domain without exit here is inconsistent as x_2 should not be s/p, but we force.

c) According to MRV, it will check-

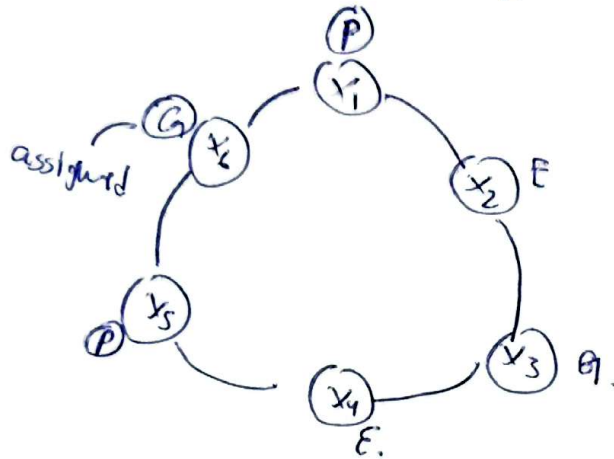
$x_1 = \text{size } 1$ and x_6 .

$x_2, x_3, x_4 = \text{size } 2$. and $x_5 = \text{size } 3$.

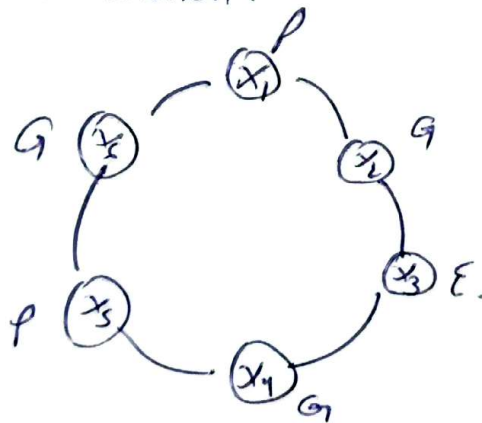
MRV would pick the smallest domain first which is x_1 and x_5 .

d) MRV would pick x_1 and x_6 first as their domain size

e) is one, then x_2, x_3, x_4 .



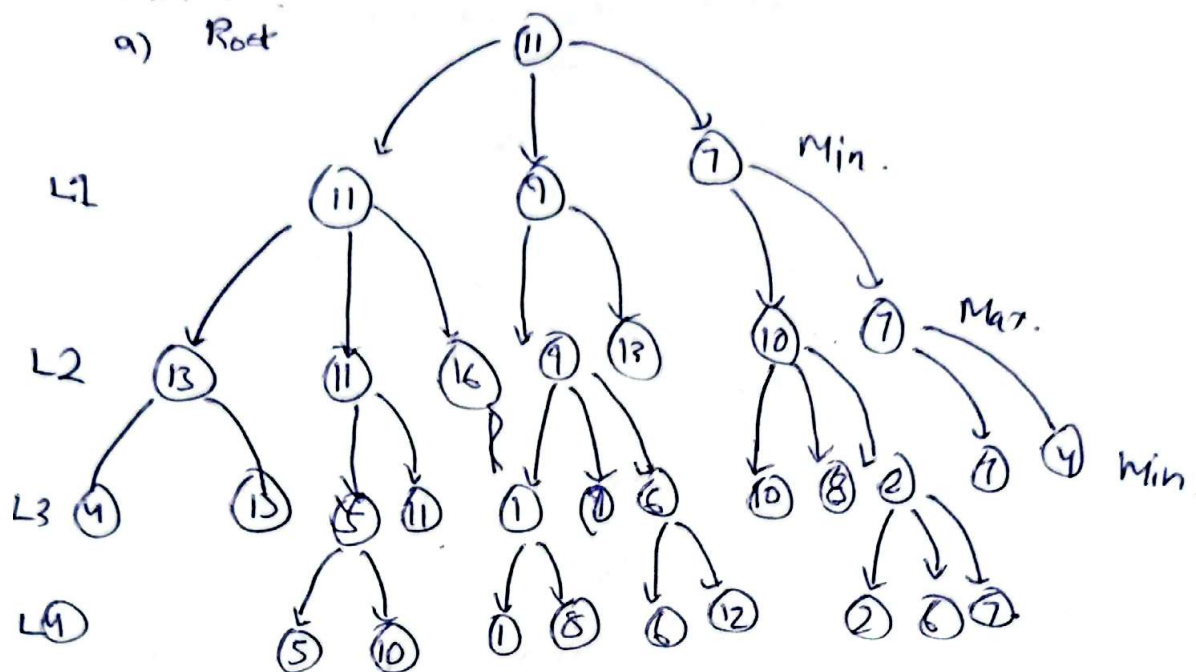
if another solution.



(yes the solution exist).

two solutions

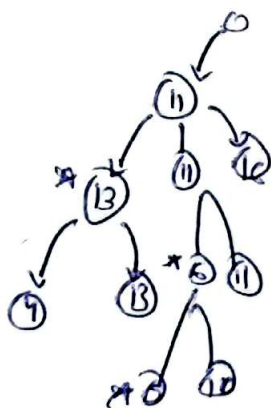
a) Root



b) Root (max)

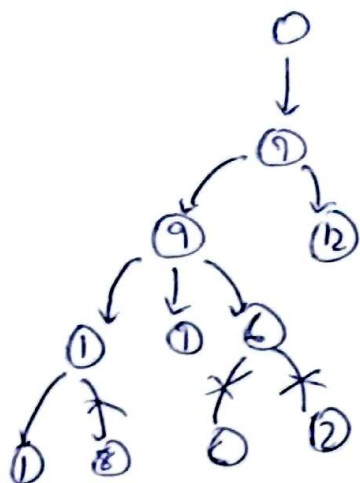
$$\alpha = -\infty \quad \text{and} \quad \beta = \infty$$

1) From the leftmost Branch.



- L3 pick 5 as min.
 - L2 pick 13 as max
and 11 as max.
 - L1 choose 11 as min.
- Root now know
 $\alpha = 11$ and $\beta = \infty$.

2) The Middle Branch.



Now at L1

$9 \leq 11$? Yes
Prune the 12.

evaluating the first leaf.
L3 would be picking 1 as min.

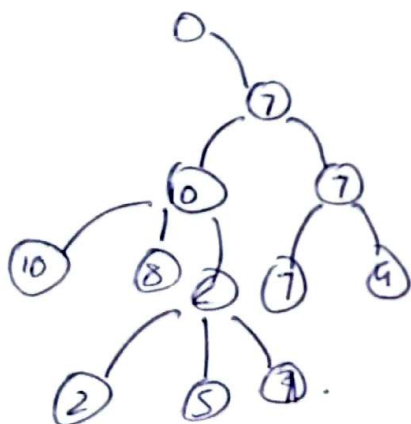
check $1 \leq 11$? Yes
Prune the leaf 8.

evaluating the leaf: 6.
check $6 \leq 11$? Yes.
Prune the leaf.

L2 choose the 9 max
from (1, 9, 6).

thus this branch gives the root 9
which is worse.

3) The Rightmost Branch.



L3 choosing leaf 2
 $2 \leq 11$? Yes
Prune 5 and 9.

L2 choosing max
10 from (10, 8, 2).

L1: evaluates 10. $10 \leq 11$? Yes
Prune everything won't give before
answer than 11.

	α	β	Result.
Top	11	α	11
Middle	9	9	Pruned.
Right	11	10	Pruned.

c) No, Alpha-beta pruning algorithm is designed to give exact same answer as the min-max algorithm. The advance part is that it just skip the branches by pruning them as their result won't change the value of root node.

Question 3:

$$a) = \frac{1}{3} \left(\frac{13}{37} + \frac{17}{51} + \frac{15}{47} \right)$$
$$= \frac{1}{3} \left(\frac{1}{3} + \frac{1}{3} + \frac{15}{47} \right)$$

$$: \frac{1}{3} \times \frac{139}{141} = \frac{139}{423}$$

b) $P(\text{Document} = 1 \mid \text{word} = \text{'we'})$,
using Bayes Theorem

$$P(D_1 | w_e) = \frac{P(w_e | D_1) \cdot P(D_1)}{P(w_e)}$$

Computing $P(w_e)$

$$\frac{1}{3} \left(\frac{1}{37} + \frac{1}{51} + \frac{2}{47} \right) = \frac{125}{5499}$$

Applying Bayes:

$$\frac{\frac{1}{37} \cdot \frac{1}{3}}{\frac{125}{5499}} = \frac{5499}{118 \times 125} = \frac{47}{125}$$

c) $P(D_2 \mid \text{word with 'a'})$

$$P(\text{am or are} | D_2) = \frac{\text{count}(\text{am}) + \text{count}(\text{are})}{\text{total } k}$$

$$P(\text{a-word} | D_2) = \frac{1}{3} \left(\frac{1}{3} + \frac{1}{3} + \frac{16}{47} \right) = \frac{142}{423}$$

Applying Bayes,

$$P(D_2 | \text{a-word}) = \frac{P(\text{a-word} | D_2) \cdot P(D_2)}{P(\text{a-word})}$$

$$= \frac{\frac{1}{3} \cdot \frac{1}{3}}{\frac{142}{423}} = \frac{47}{142}$$

$$d) P(\text{word} = \text{grat}) \\ = \frac{13}{37} \cdot \frac{1}{6} + \frac{17}{51} \cdot \frac{1}{3} + \frac{15}{47} \cdot \frac{1}{2}$$

convert to common denom (246)

$$\frac{47}{346} + \frac{474}{346} + \frac{135}{346} = \frac{46}{144}$$

$$e) P(\text{doc} = 1 | \text{word} = \text{we})$$

$$P(\text{we}) = \frac{1}{37} \cdot \frac{1}{6} + \frac{0}{51} \cdot \frac{1}{3} + \frac{2}{47} \cdot \frac{1}{2} = \frac{1}{234} + \frac{1}{47}$$

convert to common denom (10998)

$$\frac{47}{10998} + \frac{234}{10} = \frac{281}{10998}$$

Bayes Theorem

$$P(D_1 | \text{we}) = \frac{P(\text{we} | D_1) \cdot P(D_1)}{P(\text{we})} = \frac{47}{281}$$

$$ii) P(\text{a-word}) = \frac{1}{2} \cdot \frac{1}{6} + \frac{1}{3} \cdot \frac{1}{3} + \frac{16}{47} \cdot \frac{1}{2} = \frac{1}{17} + \frac{1}{9} + \frac{8}{47}$$

convert to common denom.

$$\frac{47}{346} + \frac{94}{846} + \frac{114}{846} = \frac{285}{846} = \frac{95}{282}$$

Applying Bayes.

$$P(D_2 | \text{a-word}) = \frac{P(\text{a-word} | D_2) \cdot P(D_2)}{P(\text{a-word})} \\ = \frac{94}{285}$$

Question 4:

a) $P(R)$?

rain depends on spring, thus we consider 2 cases.

Spring	$P(S)$	$P(R S)$	Product
True	0.25	0.7	0.175
False	0.75	0.3	0.225

$$\text{Sum up} = 0.175 + 0.225 = 0.4$$

b) $P(\text{wet})$

wet grass depends on worms & rains from (a) part: $P(R) = 0.4$
and $P(\neg R) = 0.6$, Total 4 combination.

Rain	worms	$P(R)$	$P(R W_0)$	$P(\text{wet} R, W_0)$
Yes	Yes	0.4	0.7	0.12
Yes	No	0.4	0.3	0.25
no	Yes	0.6	0.2	0.10
no	No	0.6	0.8	0.08

$$\text{Result} = 0.0336 + 0.03 + 0.012 + 0.0384 = 0.1140$$

c) Given, is spring already thus we are ought to use $P(R|S)$ or $P(\neg R|S)$, Other side morning depends on wet, depend on rain and worm.

Rain	Worm	$P(R S) \times P(W_0 R)$	$P(M \text{ through we})$
Y	Y	0.49	0.372
Y	N	0.21	0.32
N	Y	0.06	0.38
N	N	0.24	0.388

$$= 0.1823 + 0.0672 + 0.0228 + 0.093 = 0.3653$$

d) $P(B|S)$
 Spring is given. Bird depends on S and worm only. Rain still affects worms. We would be looping over $r \times w$.

Rain	Worms	$P(R S) \times P(W R)$	$P(B S, W)$
Y	Y	0.49	0.8
Y	N	0.21	0.9
N	Y	0.06	0.8
N	N	0.24	0.4

$$\text{Result} = 0.392 + 0.084 + 0.048 + 0.096 \\ = 0.62$$

e) $P(S|B, \neg W)$
 We know $B = \text{yes}$ & $\neg W = \text{No}$ we want the prob of $S = \text{yes}$

Rain	$P(R S)$	$P(\neg W R)$	sub total
Yes	0.7	0.3	0.21
No	0.3	0.8	0.24

$$\text{Sum} = 0.450$$

Rain	$P(R, \neg B)$	$P(\neg W R)$	sub total
Y	0.3	0.3	0.09
N	0.7	0.8	0.56

$$\text{Sum} = 0.1950$$

Normalizing

$$\text{Total}(Z) = 0.45 + 0.1950 \\ = 0.2400$$

$$\text{Answer} = \frac{0.045}{0.24}$$

$$P(S|B, \neg W) = 0.1875$$

Question 5:

A) Since this is an open-source filter built for the general public, a prior of 0.1 is sensible because it assumes most users only receive 10% spam mails, so that it avoids accidentally blocking legit emails.

Add-on smoothing

$$\frac{\text{count of word} + 1}{\text{number of emails} + 2}$$

Word	Regular	Spam
'study'	$\frac{4}{6} = \frac{2}{3}$	$\frac{1}{10}$
'free'	$\frac{2}{6} = \frac{1}{3}$	$\frac{9}{10}$
'money'	$\frac{2}{6} = \frac{1}{3}$	$\frac{5}{10}$

b) $P(\text{spam} / \text{'money' or 'Psychology study'})$

Identify the words

$$\text{money} = 1, \text{free} = 0, \text{study} = 1$$

using naive Bayes (spam score)

$$P(\text{spam}) = P(\text{study} | \text{spam}) \cdot P(\text{money} | \text{spam}) \cdot P(\text{free})$$

$$0.1 = \frac{1}{10} \cdot \frac{1}{10} \cdot (1 - \frac{9}{10}) = \frac{1}{2000}$$

(Regular source)

$$P(\text{reg}) = P(\text{study} | \text{reg}) \cdot P(\text{money} | \text{reg}) \cdot P(\text{free} | \text{reg})$$

$$(1 - 0.1) \cdot \frac{2}{3} \cdot (1 - \frac{1}{3}) \cdot \frac{1}{3} = \frac{2}{15}$$

Normalizing

$$P(\text{spam} | S) = \frac{\frac{1}{2000}}{\frac{1}{2000} + \frac{2}{15}} = 0.00327 \approx 0.37\%$$

This is a regular email. Since it contains study and money but no 'free' this word is present in almost all spam mails.

$$c) P(\text{spam}/s) > 0.5$$

For a 50/50 calculation, the spam & regular scores must be equal

$$P: P(s/\text{spam}) = (1-p) \times P(s/\text{regular})$$

$$1 \times \frac{1}{200} = (1-p) \times \frac{4}{27}$$

$$27p = 800(1-p)$$

$$827p = 800$$

$$p = 0.967$$

97% of all emails are spam before this sentence gets a 50/50 chance of all spam, just because of absence of 'free' word.

Question 6.

crabs (x-intercept) and clams (y-intercept)

$$y = mx + b$$

(m) slope and b (intercept).

Step 1: Finding the avg (data)

$$\text{total crabs} = 29$$

$$\text{total clams} = 1249$$

$$xy = 274 + 280 + 368 + 0 + 140 + 0 + 366 + 5 + 253 + 950 = 2136$$

$$x^2 = 4 + 16 + 4 + 25 + 16 + 1 + 25 + 1 + 9 = 109$$

$$\text{Total number of cages} = 10$$

Step 2: Finding slope.

$$m = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

$$= \frac{10(2136) - (29)(1249)}{10(109) - (29)^2}$$

- This means that adding one extra crab would result in approx 60 clam decrease.

- Number (-ve) because more crabs,
- fewer clams

Step 3: Calculating the b)

$$b = (\text{avg of } y) - m \cdot (\text{avg of } x)$$

$$\text{avg of } (x) = 2.9$$

$$\text{avg of } y = 124.9$$

$$b = 124.9 - (-59.88) \cdot 2.9$$

$$b = 298$$

If there are zero crabs, we expect to see around 298 clams surviving, which makes sense as we are starting from 300.

Final Answer

$$y = -59.6x + 298 \quad / \quad y = -60x + 298$$

Question 7:

Logistic Regression Formula.

$$P = \frac{1}{1 + e^{-x}}$$

where x is $b_0 + b_1x_1 + b_2x_2 + \dots + b_nx_n$

in simplified $b_0 + b_1 \text{age} + b_2 \cdot \text{diabetes} + b_3 \cdot \text{hypertension} + b_4 \cdot \text{smoker} + b_5 \cdot \text{Surgery Type}$.

New Patient = age(60), diabetes(1), hypertension(1) + smoker(1), Surgery

Since logistic regression return probability based on the coefficients ($b_0, b_1, b_2, \dots$) and they are not given in the question, the exact probability could not be predicted thus in just giving it a random values)

from the data set complication = 1 when age is high, diabetes = 1, sugar > 2, is not a smoker & is hypertensive.

Assigning random values

$$x = -5 + 0.05(\text{age}) + 1(\text{diabetes}) + 0.8(\text{hp}) + 0.7(\text{smoker}) + 1.2(\text{Surgery})$$

Add patient values

$$x = -5 + 0.05(60) + 1(1) + 0.8(1) + 0.7(1) + 1.2(12)$$

$$x = 2.9$$

$$P = \frac{1}{1 + e^{-2.9}}$$

$$P \approx 0.95$$

The patient has a very high chance 95% of complications

close to 1 $\rightarrow$ high Risk

close to 0 $\rightarrow$ low Risk.